

Day 5 – Extensions to Single Cell Analyses (2hrs)

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Objective: This workshop is designed to provide participants with a foundational understanding of the strategic decisions and computational workflows involved in single-cell RNA-sequencing (scRNA-seq) data analysis. The session emphasizes not only the practical implementation of a standard analysis pipeline but also the critical evaluation of its underlying assumptions and limitations.

Prerequisites and Environment Setup

Goal: To establish a standardized computational environment and provide the necessary data and context, ensuring participants can engage directly with the analytical content during the workshop.

- **Provided Dataset and Experimental Context:**
 - **Methodology:** A well-characterized public dataset of ~3,000 human Peripheral Blood Mononuclear Cells (PBMCs) from 10x Genomics will be used. This selection allows the session to focus on analytical methodology rather than complex biological interpretation. To demonstrate differential gene expression, a simulated case-control experimental design has been programmatically introduced into the dataset's metadata. This pedagogical approach facilitates instruction on statistical contrasts within a single, self-contained dataset.
 - **Materials:** A directory containing the gene expression count matrix (Matrix Market format) and a metadata file.
- **Technical Setup Guide:**
 - **Methodology:** A document will be provided that details the required software stack (R and RStudio) and the installation of specific R packages from CRAN and Bioconductor (e.g., [Seurat](#), [tidyverse](#)). This pre-session setup is designed to maximize the time dedicated to instruction and hands-on analysis.
 - **Materials:** A markdown file guide with installation commands and verification steps.

Core Analytical Workflow

Goal: To systematically guide participants through the standard scRNA-seq analysis workflow, pairing theoretical concepts with practical application and highlighting critical methodological considerations.

- **Didactic Component: Theoretical Foundations:**

- **Methodology:** A slide-based presentation will establish the conceptual framework for the session. Topics will include: the transition from bulk to single-cell transcriptomics, an overview of the analytical pipeline, the rationale for quality control, the principles of dimensionality reduction and clustering, and the strategic differences between primary analytical approaches.
- **Materials:** Presentation slides and jupyter notebooks.
- **Practical Component: Annotated R Notebook:**
 - **Methodology:** The core of the workshop will be a guided session using an R Notebook. The analysis is partitioned into logical sections, each addressing a key strategic objective.
 - **Sections:**
 1. **Data Quality Control and Filtering:** Application of standard metrics (e.g., feature counts, mitochondrial gene content) to identify and remove low-quality cellular profiles. The emphasis is on data-driven heuristic selection.
 2. **Cell Clustering and Annotation:** Unsupervised clustering to partition cells into transcriptionally distinct groups, followed by biological interpretation.
 - **Methodological Consideration 1: Statistical Inflation in Marker Gene Identification:** It will be explicitly noted that using the same dataset to define clusters and subsequently perform differential expression to identify marker genes can lead to inflated statistical significance (i.e., "double-dipping"). The distinction between exploratory analysis and a statistically rigorous validation framework will be discussed.
 - **Methodological Consideration 2: Manual vs. Automated Cell Type Annotation:** The workshop will implement manual annotation using canonical marker genes to build biological intuition. Following this, a brief overview of automated annotation methods (e.g., [SingleR](#), [Azimuth](#)) will be provided, contextualizing them as scalable alternatives that leverage external reference atlases.
 3. **Differential Gene Expression (DGE) Analysis:** A comparative analysis of the two primary strategies for DGE testing between the simulated conditions.
 - **Strategy 1: Pseudo-bulk Aggregation:** This approach, wherein counts from all cells of a given type within each sample are aggregated, will be presented as a statistically robust method. It appropriately models the sample as the unit of replication, leveraging well-established statistical tools from bulk RNA-seq.
 - **Strategy 2: Cell-Specific Contrasts and Associated Caveats:** This approach performs tests directly on the disaggregated single-cell data. A critical caveat will be discussed: **the issue of pseudo-replication**, where individual cells are incorrectly treated

as independent biological replicates. This violates statistical assumptions and can lead to a severely inflated false-positive rate. This discussion will reinforce the statistical rigor of the pseudo-bulk approach for most experimental designs.